PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GORVERNMENT

LONGIDO DISTRICT COUNCIL

NAMANGA SECONDARY SCHOOL

FORM THREE MID TERM EXAMINATION.

CODE: 032/1 **CHEMISTRY 1**

TIME: 3:00 HRS **Monday, 28TH AUGUST 2023, AM**

INSTRUCTIONS

i. This paper consists of section A, B and C with a total of thirteen (13) questions.

ii. Answer **ALL** in section A and B and only two questions from section C.

iii. The following constants may be used

Atomic masses:H=1, C=12, N=14, O=16, Na=23, Cu=64, Zn= 65, Pb= 207, Ag=108, Cl=35.5

Avogadro's number=6.02×10^{23}

G.M. V=$22.4dm^3$

SECTION A(16marks)

1. For each of items (i)-(x), choose the correct answer from the given alternatives and write its letter beside the item number in the answer sheets provided.

i. What is the maximum number of electrons in the innermost shell of an atom?

A. 3
B. 1
C. 4
D. 2
E. 8

ii. What feature is essential for a good fuel?

A. High speed of continuous energy supply
B. High energy value supplied
C. Low carbon dioxide supplied
D. High content of non-combustible material
E. Very high ignition point

iii. What conclusion can be drawn from the random movement of pollen grains suspended in air?

A. Matter is lighter in nature
B. Matter is solid in nature
C. Matter is particulate in nature
D. Matter is wave in nature
E. Matter has no particles

iv. Which of the following is not a component of the First Aid Kit?

A. Goggles
B. A pair of scissors
C. Knife
D. Gloves

E. Razor blade

v. Which element is oxidized in the following reaction?

$2FeSO_4 + Cl_2 + H_2SO_4 \longrightarrow Fe(SO4)_3 + 2HCl.$

A. Chlorine
B. Hydrogen
C. Oxygen
D. Sulphur
E. Iron

vi. Which of the following are the components needed to start fire?

A. Match box, fire wood and kerosene
B. Match box, fire wood and oxygen
C. Oxygen, fuel and fire wood
D. Oxygen, heat and match box
E. Oxygen, fuel and heat

vii. The ions that are removed when writing ionic equations are called...

A. active ions
B. spectator ions
C. cations
D. anions
E. reactive ions

viii. A simple proof that chemical reactions is taking place in our bodies is that,

A. We eat a balanced diet
B. Doctors tell us in the hospital
C. We occasionally fall sick
D. The food we eat and liquids we drink differ from waste products we excrete
E. We were created so by God

ix. The subatomic particles which are not found in the nucleus of an atom are,

A. Protons
B. Electrons
C. Neutrons
D. Nucleons
E. Protons and Neutrons

x. The solid compound Q was put on a watch glass and left in air. After 40 minutes the compound was found to have water but not form a solution. This compound is said to be;

A. Soluble B. Deliquescent C. Hydroscopic D. Efflorescent
E. Water used was temporary hard.

2. Match the phrase in list A with responses in list B by writing the letter of the correct response below the corresponding item number in a table provided.

LIST A		LIST B
i.	Covering small wounds	A. Liniment
ii.	Covering wounds to protect them from dirty and germs	B. Whistle C. Cotton wool D. Bandages
iii.	Cleaning wounds to kill germs	E. Safety pins F. Scissors or razor blades
iv.	Cleaning and drying wounds	G. Antiseptic H. Plaster
v.	Soothing chapped skin	I. Sterile gauze
vi.	Informing people for quick help.	J. Soap K. Petroleum jelly

SECTION B (54marks)

Answer **ALL** questions in this section, each question carries **9marks**

3. a. You are provided with a solution in bottle labelled A, B, C and D attached with P^H of 5, 6, 13 and 7respectively . What will be the effect on the litmus paper when dipped into the given solutions? **Fill the table below** **(8marks)**

SOLUTION	COLOUR ON RED LITMUS PAPER	COLOUR ON BLUE LITMUS PAPER
A		
B		
C		
D		

b. What is a fuel ? **(1marks)**

4. a. What is volumetric analysis? **(3marks)**
   b. What volume of hydrochloric acid with 37.1% purity and density of 1.2g/cm^3 should be used to prepare 1litre of 0.1000M of HCl solution. **(6marks)**
5. a. What is chemical equilibrium? **(3marks)**
   b. Read the table below and answer the questions that follows

experiment number	first reactant	second reactant	product	condition of the wall of the beaker containing mixtures	number of moles of first reactant	number of moles of second reactant	number of moles of product	enthalpy change
1	A	B	M	Hot	4	2	2	46
2	C	D	N	Cold	1	3	4	29
3	E	L	P	Hot	2	1	3	76
4	F	G	R	cold	3	2	2	88

Questions

i. Write the balanced reversible reactions of experiment 1 and 4 by showing the enthalpy changes. **(2marks)**
ii. identify experiments which are exothermic and which are endothermic. **(2marks)**
iii. Identify the experiments whose position of equilibrium can be affected by pressure. **(2marks)**

6. a. mention two merits of water hardness **(3marks)**
   b. Describe any three methods of softening hard water **(6marks)**
7. a. Define the term chemical equation **(3marks)**
   b. Complete and balance the following chemical reactions **(6marks)**
   i. $MCO_{3(S)} + HCl(aq) \longrightarrow$
   ii. $MgCl + AgNO_3 \longrightarrow$
   iii $CaO(s) + H_2O(l) \longrightarrow$
8. a. Define stoichiometry **(2marks)**
   b. How many moles of nitrogen gas would react with excess hydrogen gas to produce 2.24dm3 of ammonia gas at STP? **(5marks)**
   c. State Avogadro's law **(2marks)**

SECTION C (30 marks)

Answer only two (2) questions from this section

9. a) State Faraday's first law of electrolysis. **(2marks)**

b) Copper (II) sulphate solution undergo electrolysis by using carbon electrodes.

i. write the chemical formula of the electrolyte used **(2marks)**
ii. mention four important ions present in the solution **(2marks)**
iii. show the reactions taking place at each electrode and identify the main product on cathode and anode. **(4marks)**
iv. write the overall ionic equation for the electrolysis above **(2marks)**
v. Draw the electrolytic cell for this electrolysis. **(3marks)**

10. a. In an experiment of investigating the effect of concentration on the rate of chemical reaction between sodium thiosulphate and dilute hydrochloric acid by observing the disappearance of cross mark, the following results were obtained.

Volume of $Na_2S_2O_3$ (cm^3)	**Volume of water(cm^3)**	**Total volume(cm^3)**	**Time, t(s) taken for the cross to disappear**	**Rate of reaction, $1/t(s^-)$**
30	0	30	0	
25	5	30	12	
20	10	30	36	
15	15	30	72	
10	20	30	152	
5	25	30	280	

i. Complete the table above (**3marks)**

ii. Write a balanced chemical equation of the experiment above **(3marks)**
iii. Write the ionic equation of the reaction above. **(3marks)**
iv. Plot a graph of volume of $Na_2S_2O_3$ against rate of reaction. **(6marks)**

11. a. What is extraction of metals? (2marks)
b. With the aid of clear labelled diagram and equations explain how sodium metal is extracted from its chief ore. **(13marks)**

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